TechNova Customer-Support Chatbot

Production Readiness, Performance, Reliability & Evaluation Report

Technology LangChain • LangGraph • NVIDIA NIM • LangSmith	**Primary benchmark model** nvidia/nemotron-3-super-120b-a12b
Evaluation dataset 12 customer-support questions	**Feedback study** 5 recorded user-feedback interactions

1. Project Overview

The TechNova chatbot is a tool-using customer-support assistant designed to answer order-status and company-policy questions. The application uses LangGraph to orchestrate an agent/tool loop, LangChain tools for deterministic business data, NVIDIA NIM-backed chat models for natural-language reasoning, and LangSmith for tracing, evaluation, token analysis and feedback collection.

Core workflow

1	**Customer message**	**User asks about an order, return, warranty or shipping.**
2	Agent node	LLM interprets the request and decides whether a tool is needed.
3	Tool node	get_order_status or get_policy returns grounded TechNova data.
4	Agent node	LLM produces the final customer-facing response.
5	LangSmith	Run tree, metadata, tokens, latency, evaluation and feedback are recorded.

Implemented support tools

get_order_status(order_id) provides status, item and expected delivery date. **get_policy(topic)** provides returns, warranty and shipping policies. The graph also uses MemorySaver with a thread ID so the conversation can retain state across turns.

Production-readiness focus: the project was evaluated for response quality, latency, token growth, cost, fault handling, model fallback, retries and observability.

2. Performance, Tokens & Cost

End-to-end LangGraph latency benchmark

A 20-question end-to-end benchmark using the full TechNova LangGraph workflow with Nemotron 120B produced an average latency of **2.85 s**, P50 of **2.70 s**, P95 of **4.80 s**, minimum 1.15 s and maximum 4.83 s. LangSmith node traces showed that the second LLM call dominated slow runs while tool execution was negligible.

Separate model-comparison benchmark

A separate model-level benchmark was used specifically to compare the two working models. Because it used a different benchmark setup, its latency figures should not be directly mixed with the end-to-end figures above.

Metric	Nemotron 120B	GPT-OSS 20B
Average latency	8.48 s	10.70 s
P50	9.55 s	9.04 s
P95	11.84 s	21.98 s
Minimum	3.07 s	1.14 s
Maximum	12.23 s	25.20 s

Interpretation: Nemotron was faster on average and substantially more stable at the tail in this particular model-comparison benchmark. GPT-OSS 20B had a slightly lower P50 but a much higher P95. This is a benchmark result, not a universal claim about the models.

TTFT and token behavior

Measurement	Observed result
TTFT average (5 streamed questions)	2.347 s
TTFT P50	2.156 s
Baseline 10-turn input tokens	28,429
Last-4-message input tokens	9,097
Input-token reduction	≈68.0%
Turn-10 input reduction	≈82.8%

20-question token/cost benchmark

Model	Input	Output	Total	Hypothetical cost
GPT-OSS 20B	1,446	10,337	11,783	$0.002356
Nemotron 120B	475	16,399	16,874	$0.015187

The cost figures use the assignment's hypothetical rate card: approximately $0.20/1M tokens for the small-model mapping and $0.90/1M tokens for the large-model mapping. They are exercise estimates rather than actual NVIDIA billing.

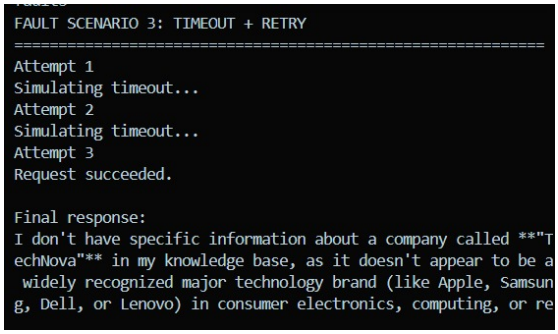
10,000 conversations/day scenario: treating one 20-question benchmark as one conversation gives estimated daily costs of $23.56 for GPT-OSS 20B and $151.87 for Nemotron 120B; 30-day estimates are $706.80 and $4,556.10.

3. Reliability, Fault Injection & Evaluation

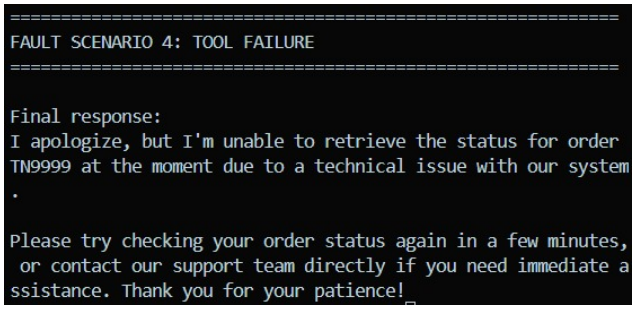
Fault-handling experiments

Five resilience scenarios were tested: invalid-model fallback, invalid API-key fail-fast behavior, timeout retry, tool failure handling, and bad-input validation. The timeout experiment succeeded on the third attempt, while tool failure produced a graceful customer-facing response instead of crashing the application.

Scenario	Observed behavior
Invalid model	Primary model failure triggered fallback to the backup model.
Invalid API key	403 authorization failure; application failed fast without retry.
Timeout	Attempts 1 and 2 timed out; attempt 3 succeeded.
Tool failure	Tool error was converted into a graceful support response.
Bad input	Empty/oversized input was rejected before the LLM call.



Evidence: timeout retry

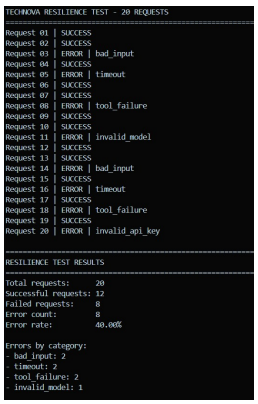


Evidence: graceful tool failure

20-request resilience test

The injected resilience benchmark produced 12 successful requests and 8 failed requests, for a measured error rate of **40.00%**. This value reflects the deliberately injected fault mix and should not be interpreted as normal production reliability.

Fault category	Count
Bad input	2
Timeout	2
Tool failure	2
Invalid model	1
Invalid API key	1



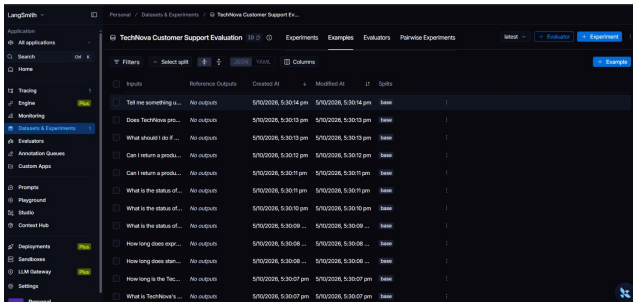
Evidence: complete 20-request resilience run, including requests 13–20 and the final error breakdown.

4. Evaluation, Recommendation & Production Risks

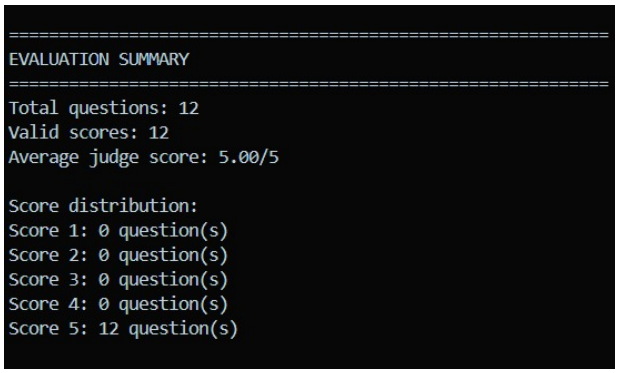
LangSmith evaluation

A LangSmith evaluation dataset containing 12 customer-support questions was created. A reference-aware LLM-as-judge evaluation was run for both benchmark models. Nemotron 120B achieved an average judge score of **5.00/5** (12 of 12 questions scored 5). GPT-OSS 20B achieved **4.67/5** (11 questions scored 5 and one scored 1). The same judge model, GPT-OSS 20B, was used for the controlled comparison.

Evaluation	Nemotron 120B	GPT-OSS 20B
Questions	12	12
Valid evaluations	12/12	12/12
Average judge score	5.00 / 5	4.67 / 5
Score 5	12	11
Lower score	0	1



Evidence: LangSmith evaluation dataset containing 12 examples.



Evidence: Nemotron evaluation summary showing 12 valid scores and a 5.00/5 average.

Recommendation

Recommended primary model: nvidia/nemotron-3-super-120b-a12b. In this controlled evaluation it achieved the higher quality score and, in the separate model-comparison benchmark, lower average latency and much better P95 stability than GPT-OSS 20B. GPT-OSS 20B remains attractive when cost is the dominant constraint because the exercise's hypothetical cost model is substantially lower.

Top 3 production risks and alert thresholds

Risk	Why it matters	Suggested alert threshold
LLM latency	Slow second LLM calls can dominate end-to-end response time.	P95 latency > 5 s
Context/token growth	Long histories increase input tokens, cost and latency.	Avg input tokens > 500
Application errors	Model/API/tool failures can affect availability.	Error rate > 5%

User feedback: five feedback interactions were successfully recorded in LangSmith covering return policy, warranty, order status, shipping and an unrelated-question refusal.

Conclusion: The TechNova chatbot demonstrates an observable LangGraph workflow with deterministic tools, memory, model fallback, retry handling, evaluation and feedback. Before production launch, priorities should be reducing tail latency, enforcing context/token controls in the main graph, and replacing the intentionally injected 40% resilience-test error rate with normal-load reliability monitoring.

Evidence note: screenshots in this report are from the executed TechNova-Bot experiments supplied for this report. Benchmark costs use the assignment's hypothetical pricing assumptions. The GPT-OSS 20B evaluation score is reported from the completed evaluation run; its summary screenshot was not included in the supplied evidence.